

Connecting Multiple MikroTik Routers

Via 1 Central Hostinger VPS (Bypassing CGNAT & Private IPs)

Production Operations & Network Engineering Manual • African Network

1. The Challenge (CGNAT & Private IPs)

When deploying MikroTik routers on client sites, cellular networks (MTN, Airtel, Glo), Starlink, or local fiber ISPs use Carrier-Grade NAT (CGNAT). The router gets a private WAN IP (e.g. 100.64.x.x or 10.x.x.x). Cloud servers (like Vercel) CANNOT initiate incoming connections to private IPs. Furthermore, MikroTik IP Cloud DDNS fails behind CGNAT because the public IP is shared and inbound ports are blocked by the ISP.

2. The Solution: 1 Single Hostinger VPS Hub

A single Hostinger Ubuntu VPS with a static public IP acts as a secure WireGuard VPN and port-forwarding hub. Each MikroTik router dials OUT to the VPS on UDP port 51820. Because the connection is outbound, CGNAT and ISP firewalls never block it. The VPS assigns a private tunnel IP (10.100.0.X) to each router and forwards a unique external port (8443, 8444, 8445...) to that router's REST API (port 443).

Router Name	Location	WireGuard IP	VPS Public Port	Router Port	Status
Router 1 (Main Gateway)	Kano Branch	10.100.0.2	8443	443 (SSL)	Primary Hub
Router 2 (Campus Wi-Fi)	Abuja Site	10.100.0.3	8444	443 (SSL)	Direct REST
Router 3 (Hotel / Café)	Lagos Site	10.100.0.4	8445	443 (SSL)	Direct REST
Router N (Any Location)	Site N	10.100.0.N	8440 + N	443 (SSL)	Unlimited

Capacity: A single \$5/mo Hostinger VPS (2GB RAM, 1 vCPU) can easily manage 50+ MikroTik routers.

3. Step 1: Initial Hostinger VPS Setup (Run Once)

SSH into your Hostinger VPS as root and execute this automated setup script:

HOSTINGER VPS - WIREGUARD HUB SETUP SCRIPT

```
# 1. Update and install WireGuard & firewall tools
apt-get update && apt-get install -y wireguard iptables-persistent ufw

# 2. Enable kernel IPv4 forwarding
sysctl -w net.ipv4.ip_forward=1
sed -i "s/#net.ipv4.ip_forward=1/net.ipv4.ip_forward=1/" /etc/sysctl.conf

# 3. Generate Server Keypair in /etc/wireguard
mkdir -p /etc/wireguard && cd /etc/wireguard && umask 077
wg genkey | tee server_private.key | wg pubkey > server_public.key

# 4. Create WireGuard Server Config (/etc/wireguard/wg0.conf)
cat <<EOF > /etc/wireguard/wg0.conf
[Interface]
Address = 10.100.0.1/24
ListenPort = 51820
PrivateKey = $(cat server_private.key)
PostUp = iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
PostDown = iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING -o eth0 -j MASQUERADE
EOF

# 5. Enable and start WireGuard daemon
systemctl enable wg-quick@wg0 && systemctl start wg-quick@wg0

# 6. Open WireGuard & Router Ports in UFW Firewall
ufw allow 22/tcp && ufw allow 51820/udp && ufw allow 8443:8499/tcp && ufw --force enable

# Print VPS Public Key (SAVE THIS VALUE!):
cat /etc/wireguard/server_public.key
```

4. Step 2: Configure MikroTik Router (In WinBox / Terminal)

Open WinBox on the router, click Terminal, and paste these commands:

MIKROTIK ROUTEROS V7 - WIREGUARD CLIENT COMMANDS

```
# 1. Create WireGuard interface on Router
/interface wireguard add name=wg-vps listen-port=13231 mtu=1420

# 2. Assign tunnel IP (Use 10.100.0.2 for Router 1, 10.100.0.3 for Router 2, etc.)
/ip address add address=10.100.0.2/24 interface=wg-vps

# 3. Add Hostinger VPS as WireGuard Peer
/interface wireguard peers add interface=wg-vps \
    endpoint-address="YOUR_VPS_PUBLIC_IP" \
    endpoint-port=51820 \
    public-key="PASTE_VPS_SERVER_PUBLIC_KEY" \
    allowed-address=10.100.0.0/24 \
    persistent-keepalive=25s

# 4. Enable RouterOS REST API on HTTPS (443)
/ip service set www-ssl port=443 disabled=no

# 5. Create Cloud API User for African Network
/user group add name=cloud-api policy=api,rest-api,read,write,test
/user add name=afrikanetwork group=cloud-api password="YourSecureRouterPassword123!"

# 6. Output Router Public Key (COPY THIS KEY!):
/interface wireguard print
```

5. Step 3: Register Router Peer on Hostinger VPS

Run this on the Hostinger VPS terminal to authorize the router and route port 8443:

HOSTINGER VPS - REGISTER PEER & PORT FORWARDING

```
# Variables for Router 1 (Adjust for Router 2, 3...)
ROUTER_PUBLIC_KEY="PASTE_ROUTER_PUBLIC_KEY_FROM_STEP_2"
ROUTER_IP="10.100.0.2"
VPS_PORT="8443"

# Authorize peer on running WireGuard interface
wg set wg0 peer "$ROUTER_PUBLIC_KEY" allowed-ips "$ROUTER_IP/32"

# Persist peer in /etc/wireguard/wg0.conf
cat <<EOF >> /etc/wireguard/wg0.conf

[Peer]
PublicKey = $ROUTER_PUBLIC_KEY
AllowedIPs = $ROUTER_IP/32
EOF

# Forward incoming VPS_PORT traffic to Router REST port 443
iptables -t nat -A PREROUTING -p tcp --dport $VPS_PORT -j DNAT --to-destination $ROUTER_IP:443
iptables -A FORWARD -p tcp -d $ROUTER_IP --dport 443 -m state --state NEW,ESTABLISHED,RELATED -j ACCEPT

# Save iptables rules so they persist across reboot
netfilter-persistent save
```

6. Step 4: Add Router in African Network Dashboard

Form Field	Value to Enter	Explanation
Router Name	Kano Main Gateway	Human-friendly label shown in dashboard
IP Address	YOUR_VPS_PUBLIC_IP	Hostinger VPS Static Public IPv4 Address
Port	8443	The external port forwarded to this router
Username	africannetwork	The dedicated cloud API user created in Step 2
Password	YourSecureRouterPassword123!	The secure password created in Step 2
Use SSL	Yes (Checked)	Uses HTTPS REST protocol (port 443)
DNS Name	asuktech.net	Hotspot login redirect domain
Hotspot Server	hotspot1	Default hotspot server identity on router

7. Verification & Health Checklist

- ' Test 1 (Ping from VPS): Run `ping 10.100.0.2` on VPS. Latency should be 20-50ms.
- ' Test 2 (REST API via Curl): Run `curl -k -u africannetwork:Pass https://VPS\_IP:8443/rest/`
- ' Test 3 (Dashboard Connection): In /super-admin -> Locations, click "Test Connection" -> OK
- ' Keepalive: "persistent-keepalive=25s" ensures the tunnel stays active 24/7 through mobile
- ' Adding Router 2: Repeat with IP 10.100.0.3, VPS Port 8444. Repeat for Router 3 with IP 10.100.0.4